

Background

Problem

- Today's transportation industry have created a substantial environmental impact due to the large amount of greenhouse gases emitted each year. Since trucks and airplanes are the most dominant method of cargo transportation, fossil fuel has become too reliable that it destroys the environment.

Solution

- To create a mode of transportation that reduces the reliance on gas and lower costs & emissions. An autonomous vehicle that delivers freight cost effectively, with low emissions, and independence of infrastructure can help to reduce environmental impacts.

Vision

- Project airship is a miniature model of a real industrial cargo blimp that will have functionalities to counter environmental problems. The design is an autonomous cargo airship that relies on helium for lift and electrical propulsion. The system will have navigation plan routes while avoiding obstacles & collisions without human intervention.

Key Requirements

Flight Control

- There are in total of 6 motors: 4 motors are used to control the height of the blimp. Once the blimp is at a certain altitude, the motors decrease their speed to allow the blimp to hover while it relies on helium. 2 motors are used to control the direction of the blimp, allowing it to move forward and perform wide turns

Architectural Design

Project Airship is a 3-tier layered system that interact and communicate together to operate the blimp.

Ground Station Layer

The purpose of the ground station layer is to provide an interface that allows users to control the blimp.

- Set waypoints
- Manually control the blimp
- Receive data output
- Displays camera output

Raspberry Pi Layer

The raspberry pi layer controls the main logic of the airship and communicates interfaces between the ground station and Arduino.

- Performs computer vision to locate landing spots
- Stores map data for navigation
- Performs actions from navigational data

Arduino Layer

The Arduino layer activates actuators & collects readings from the sensors which streams to the raspberry pi layer.

- Controls motor speed
- Collects data from lidar, IMU, and current sensors
- Measures battery percentage

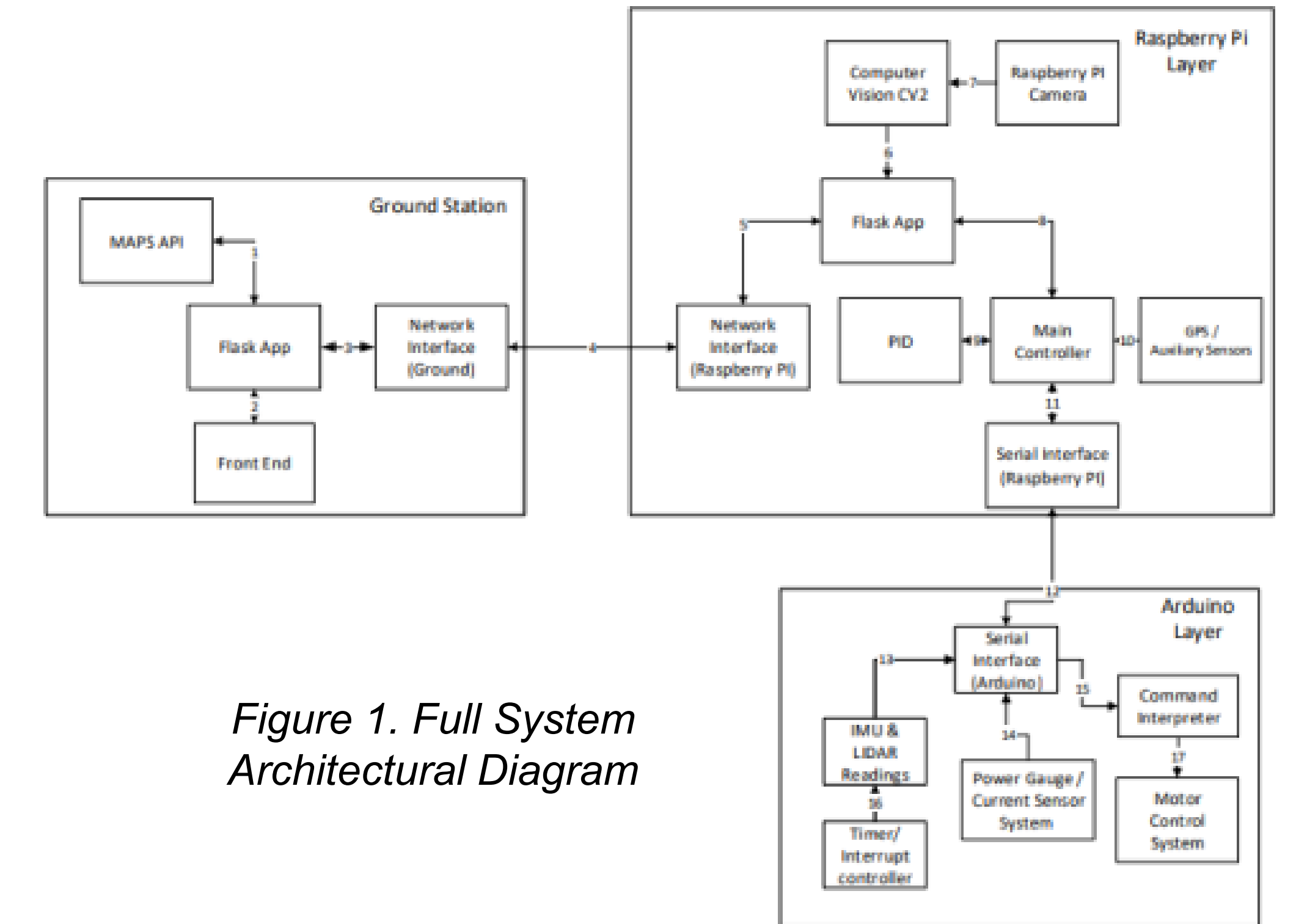


Figure 1. Full System Architectural Diagram

Implementation Details and Test Plan

Table 1. Assembly of each component

No.	Component	Description
1	Gondola	The physical structure that holds all hardware components created using thin wooden sheets & wooden rods
2	Carbon fiber tubes	Tubes are drilled to the base of the gondola
3	Landing Struts	Struts are drilled on the bottom of the gondola
4	Battery Holder	Attached to the bottom of the gondola where battery is secured and can be inserted & removed with ease
5	Motors	Motors are mounted on a custom 3D print that slides onto the carbon fiber tubes
6	ESCs	Connected with the motors and are plugged in through different pins on the Arduino
7	Raspberry Pi	Attached to the ceiling of the gondola with Velcro for easy accessibility
8	Arduino	Attached to the front of the gondola with Velcro for easy connection with other components
9	Ultra Sonic Sensor	Secured to the front right of the gondola with Velcro that connects with the Arduino
10	Battery Sensor	Secured to the front of the gondola using Velcro that connects with the Raspberry Pi

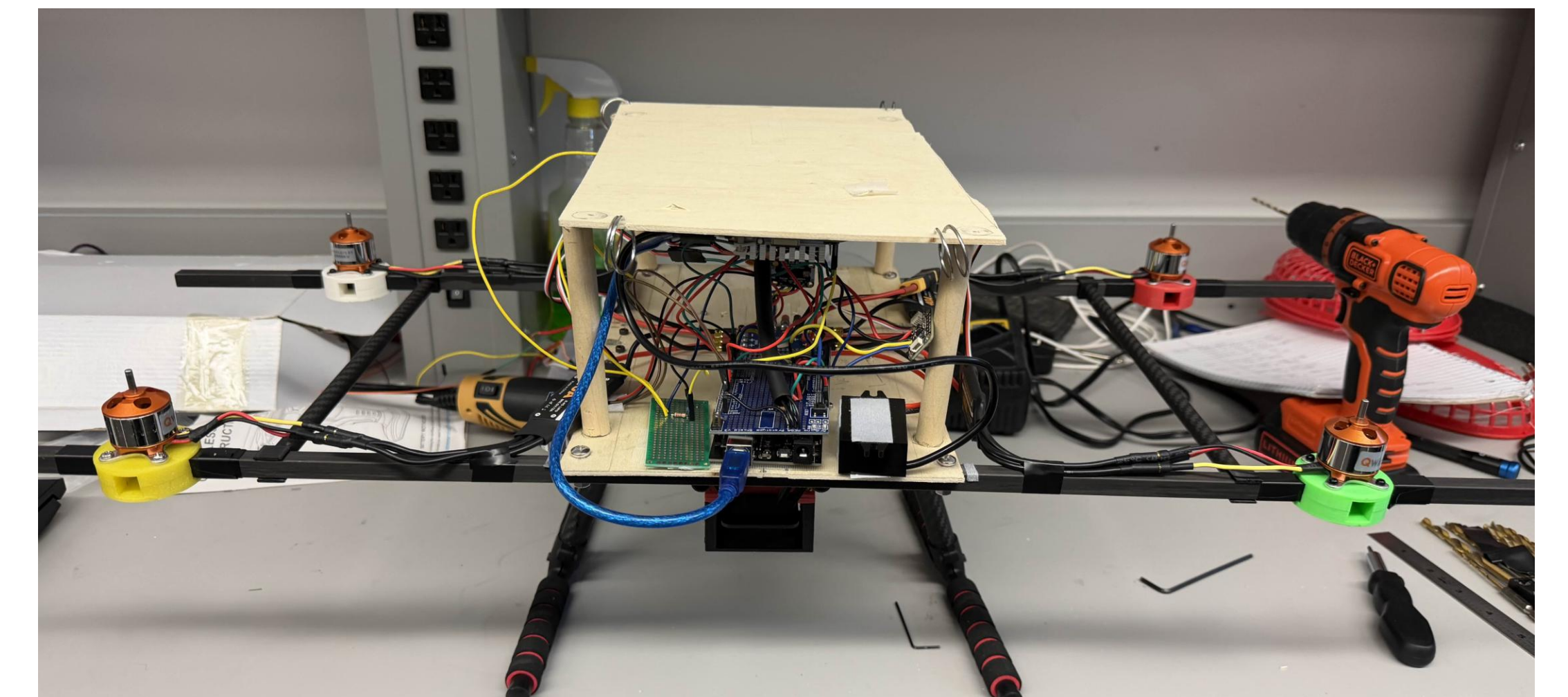


Figure 2. Construction and setup of the gondola for the blimp

Conclusions and Future Work

- Conclusion: Currently, all hardware components are communicating efficiently with minimal delays. Motors activate on command and change speed depending on altitude, Weight distribution is still off as the blimp tilts as it's trying to get off the ground.

- Future works
 - Automatic flight control
 - Implementing emergency protocol
 - Automatic take-off and landing
 - Obstacle detection

References

- Liu, S., and Bobrow, J. E., "An Analysis of a Pneumatic Servo System and Its Application to a Computer-Controlled Robot," *ASME Journal of Dynamic Systems, Measurement, and Control*, 1988, Vol 110 pp 228-235.